Round 7 — the whole error is three numbers per neuron. How do we compute them better, cheaply?

Technical memo · bias-free ReLU MLP, n=1024, depth 16, W~N(0,2/n), x~N(0,I) · 1 u = 2^{31} FLOPs = one dense n^3 product · budget = 2^{41} FLOPs = 1024 u

Executive summary. The channel-oracle experiments prove the entire final-layer error lives in three per-neuron pre-activation numbers: variance (worth −47%), κ_4 diagonal (−30%), κ_3 diagonal (−14%). I show that every pre-activation at every layer is a **Gaussian mixture** over the network weights when conditioned on the previous post-ReLU layer — because each layer draws fresh weights independent of the input — and this law (i) regenerates the κ_4 at every ReLU (matching the observed Mehler-star birth and explaining why transported stars die), (ii) predicts the per-neuron pattern lives in a *single scalar field* per layer, namely the law of $|h|^2$, and (iii) makes a **control-variate correction sampler** built on the chain's own Gaussian surrogate the cheap, exact fix for all three numbers at once. Estimated cost ≈ 16–160 u (score 0.016–0.16×budget), i.e. comparable to or below leaders while removing nearly all of the raw-MSE gap.

0. A structural fact that unifies the three channels

Write the pre-activation at layer ℓ+1 as a linear function of the previous post-ReLU layer:

$$z_i'^{(\ell+1)} = \sum_j W_{ij}^{(\ell+1)} h_j^{(\ell)}, \qquad h^{(\ell)} = \mathrm{relu}(z^{(\ell)}).$$

Because network weights at different layers are independent draws (fresh W at each layer) and the input is Gaussian, we have the **key identity**: conditional on the driving field $h^{(\ell)}$, the new weights are still i.i.d. Gaussians independent of $h^{(\ell)}$, so

$$z_i'^{(\ell+1)} \,\big|\, h^{(\ell)} \;\sim\; \mathcal{N}\big(0,\ \tilde\sigma^2\big), \qquad \tilde\sigma^2 \;=\; \frac{2}{n}\sum_j \big(h_j^{(\ell)}\big)^2 \;=\; \frac{2}{n}\,|h^{(\ell)}|^2,$$

and the neurons $i = 1..n$ of the next layer are i.i.d. given $h^{(\ell)}$. This is the standard Gaussian-mixture / GAMP recurrence. Its per-neuron marginal cumulants are, **exactly**:

$$\mathrm{Cov}(z_i') = \mathbb{E}[\tilde\sigma^2] = \frac{2}{n}\,\mathbb{E}|h^{(\ell)}|^2, \qquad \kappa_4(z_i') = 3\,\mathrm{Var}(\tilde\sigma^2) = 3\,\frac{4}{n^2}\,\mathrm{Var}\big(|h^{(\ell)}|^2\big), \qquad \kappa_3(z_i') = 0.$$

The last equality, $\kappa_3(z_i') \equiv 0$, is a **hard, cheap-to-check prediction**: the marginal skew of every pre-activation is exactly zero because a symmetric mixture of centered Gaussians is symmetric. The chain currently carries a nonzero marginal κ_{3iii} via its sum-over-sources transport; the truth the oracle injects is (near) zero. That is exactly why the κ_3-diagonal oracle helps (−14%): we are **removing a correctly-transported but exactly-zero quantity**. The nonzero two-index object the chain measures with slope 1.00, $\kappa_3(i,i,j)$ for i≠j, is *not* the marginal κ_{3iii}; it is a cross-correlation term and does not enter the per-neuron Edgeworth mean closure.

Falsifier 0 (run first, ~minutes on stored arrays). For each layer ℓ, regress the stored true marginal third sample cumulant $\kappa_3(i,i,i)$ against the chain's propagated κ_{3iii}. The Gaussian-mixture theory predicts the **truth ≈**

0 (slope / intercept consistent with 0 at MC noise), and the chain's value is the model error to be removed. Also check the level identity $\frac{2}{n}\sum_j (C_{jj} + \mu_j^2)$ reproduces the stored next-layer variance diagonal to better than 0.1%.

Two consequences worth stating before the questions.

- **κ_4 regenerates at every ReLU** from the scalar field $\mathrm{Var}(|h|^2)$ — not transported from old births. This matches item 5: a transported Mehler star contributes only ~10% of the deep diagonal, because the deep law is re-drawn each layer from the current $|h^{(\ell)}|^2$.
- **The level is flat, the pattern is finite-n.** The mixture level $\frac{2}{n}\mathbb{E}|h|^2$ is the *same* for every neuron (mean-field). The observed per-neuron pattern is the O(1/√n) fluctuation of the quadratic form $w_i^\top C_{\text{post}} w_i$ driven by the *actual* row w_i — which is exactly where the chain's covariance error projects out (item 3). So the pattern is finite-n structure on top of an exactly-flat mixture level.

1. Q1 — the variance channel (worth −47%)

Objective: the ~300–400 neurons with |μ/σ|<0.75, to ~0.1% relative (≈ 1e−3 absolute on values of order 1). We need $v_i = w_i^\top C_{\text{post}} w_i$. Three observations.

1a. It is nearly degenerate — use the mixture level plus a correction.

For a random row $w_i \sim \mathcal{N}(0, \frac{2}{n} I)$, across neurons, $\mathbb{E}[v_i] = \frac{2}{n}\,\mathrm{tr}(C_{\text{post}})$ and $\mathrm{Var}(v_i) \approx \frac{8}{n}\,\mathrm{tr}(C_{\text{post}}^2)/n \cdot \ldots$ — specifically $\mathrm{Var}(v_i) = \frac{8}{n^2}\mathrm{tr}\big(C_{\text{post}}^2\big) \approx 8/n$, so the rms spread of v_i about its mean is $\approx \sqrt{8/1024} \approx 0.09$, i.e. ~9% of a unit value. The chain's current error (0.15% relative) is already 60× smaller than the across-neuron spread, so *there is no hard per-neuron signal to recover; the problem is purely a tiny additive projection error* $\mathrm{d}v_i = w_i^\top \mathrm{d}C_{\text{post}} w_i$.

1b. Kill the projection error by sampling the correction, not the full covariance.

Split $v_i = v_i^{\text{chain}} + \mathrm{d}v_i$. The chain value is free. We estimate only the residual $\mathrm{d}v_i$, which has rms ~1.5e−3 and is a centred quantity — ideal for Monte-Carlo with a control variate. Using the Gaussian-mixture structure, the clean per-neuron target is the second moment of the next pre-activation:

$$\mathbb{E}\big[(z_i')^2\big] = \mathbb{E}\Big[\big(w_i^\top h^{(\ell)}\big)^2\Big] \quad \text{is our target;} \quad v_i^{\text{chain}} = \mathbb{E}_g\big[(z_i')^2\big] \text{ is the control.}$$

The control variate is the *analytic* Gaussian-surrogate expectation E_g, which already has 99.85% of the mass. We draw input samples $\{x^{(s)}\}_{s=1}^N$, record the driving field, and accumulate the scalar residual $\big(w_i^\top h^{(\ell)}(x^{(s)})\big)^2 - \mathit{//} E_g[(z_i')^2] \mathit{//}$. Because the residual has tiny variance, the number of samples is small.

Sample count. Per-neuron residual rms $\approx 1.5\mathrm{e}{-3}$. The Monte-Carlo mean of N samples has std $1.5\mathrm{e}{-3}/\sqrt{N}$. To reach ≤ 5e−4 (0.05% of 1), need $N \geq (1.5/5)^2 \approx 9$. Taking $N = 10^3$ gives std $1.5\mathrm{e}{-3}/\sqrt{1000} \approx 5\mathrm{e}{-5}$, a 30× margin below target; $N = 10^4$ gives 1.5e−5. One batch of N forward passes serves all active neurons and all layers simultaneously.

Cost. One forward pass ≈ 0.016 u. The estimator is one set of N forward passes — which already visits every layer, so all layers' corrections come from the same samples. Cost = $0.016\,N$ u. For $N = 2{\times}10^3$: `32 u`

`(0.031×budget)`. For $N = 10^4$: `160 u (0.156×budget)`. Both sit at/below the leaders' cost bracket while eliminating the −47% channel.

1c. The "fresh ~4e−5 per entry" off-diagonal covariance error.

Item 3 says the post-ReLU off-diagonal C has a random ~4.2e−5 rms fresh component per entry (~2% of entry size) unexplained by the chain's own closure (only 3–8% explained). The Gaussian-mixture picture explains *why it is unpredictable from chain quantities*: it is the finite-n weight-sampling fluctuation of the cross terms $\mathbb{E}[h_j h_k]$, which under the mixture law are $\mathbb{E}\left[\sigma^2 \rho_{jk}\right]$ with ρ_{jk} the (nontrivial, correlated) joint active/silent indicator of the pair. This pairwise joint information is genuinely *not in* the chain's single-neuron closures — it is the (i,j)-bivariate law, which the chain's Wick/Edgeworth program approximates but does not carry. **Conclusion:** do not chase a new closure term for this; it is the correct locus for the sampler, not for a correction formula.

Proposal 1 summary. (a) State: the active-row mask A (~350 neurons), and per-sample accumulators — size O(n+N) scalars. (b) FLOPs in n³ per layer: $O(n^2)$ for the record-and-accumulate during the forward pass ≈ `n³/1024`, negligible. (c) Falsifier: take stored true $C^{\text{true}}_{\text{post}}$, compute $v_i^{\text{true}} = w_i^\top C^{\text{true}} w_i$ on A, compare estimator against it; report max rel error and R². (d) End-to-end: feed corrected var into the mean closure every layer, remeasure final MSE; oracle bound says reaching true var drops ratio to ~0.53, i.e. final MSE ~1.22e−8.

2. Q2 — the κ₄ diagonal (worth −30%)

We need the per-neuron $\kappa_4(z'_i)$ of the next layer. The (2,2) and (1,1,1,1)-style contractions set the level; the hard part is the (2,1,1)/(1,1,1,1) per-neuron *pattern*. The mixture law gives closed forms for *all* parts with a single scalar coefficient.

2a. The regenerated κ₄ tensor at each layer.

Under the mixture law, the post-ReLU κ₄ of the pre-activation at layer ℓ+1 is determined by $\mathrm{Var}(\tilde{\sigma}^2)$. For four indices (a,b,c,d):

$$\kappa_4\left(z'_a, z'_b, z'_c, z'_d\right) \;=\; \mathrm{Var}\left(\tilde{\sigma}^2\right) \;\cdot\; \left[\,\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}\,\right], \qquad \tilde{\sigma}^2 = \frac{2}{n}|h^{(\ell)}|^2.$$

This is born fresh each layer from the law of $|h^{(\ell)}|^2$ and is a **complete pairing-sum (rank-1 "star") pattern** — exactly the observation that a Mehler star birth law is right at layer 2. The individual slots you care about:

$$\text{per-neuron diag: } \kappa_4(z'_i) = 3\,\mathrm{Var}\left(\tilde{\sigma}^2\right), \qquad (2,2)\text{ level: } \kappa_4(z'_i, z'_i, z'_j, z'_j) = \mathrm{Var}\left(\tilde{\sigma}^2\right) \;\; (i \neq j).$$

The full tensor needs only one scalar per layer, $\mathrm{Var}(\tilde{\sigma}^2) = \frac{4}{n^2}\mathrm{Var}(|h^{(\ell)}|^2)$, so **no tensor is ever materialised** — the (2,1,1) and (1,1,1,1) content is the same coefficient times a known combinatorial pattern.

2b. Evaluating the contraction w_i-by-w_i, and the per-neuron pattern.

The contraction into the next layer needs $\mathbb{E}[\kappa_4$-transport] and the finite-n pattern coming from the actual row w_i. The *level* piece is the flat mixture coefficient transported exactly. The *pattern* piece is the O(1/√n) fluctuation of the same object seen through the real weight row. Concretely, the per-neuron diagonal of the transported κ_4 is

$$\kappa_4(z_i'')\Big|_{\text{pattern}} \;=\; 3\,\frac{4}{n^2}\;\sum_{j,k}\Big[\,\mathbb{E}\big[(w_i)_j^2(w_i)_k^2\,(h_j^2-\mathbb{E}h_j^2)(h_k^2-\mathbb{E}h_k^2)\big]\,\Big],$$

which is a **single quadratic form in the law of $|\mathbf{h}|^2$** — a scalar per neuron, evaluated row-by-row. Sparse-row evaluation: since the weight moments $(w_i)_j^2 \sim \frac{2}{n}$ are factorised and h_j^2 are the previous-layer fields, the sum reduces to w-independent layer statistics plus an O(1) per-row weight factor. **This is why no full κ_4 tensor is needed.**

Falsifier on the stored arrays: we have stored $\kappa_4(i,i,j,j)$, $\kappa_4(i,i,i,j)$, and the κ_4 diagonal true tensors. The star law predicts $\kappa_4(i,i,j,j)$ is *constant in (i,j)* at fixed level (check flatness/slope), and that $\kappa_4(i,i,i,j) \propto \delta_{ij}$. Regress the stored (2,2) and (1,1,1,1) slices against the coefficient $\mathrm{Var}(\tilde{\sigma}^2)$ computed from the stored post-ReLU h distribution.

FLOP count. Per layer the contraction is O(n²) (one vector operation over the active rows), `≈ n³/1024 per layer`. The only nontrivial cost is estimating $\mathrm{Var}(|h|^2)$, which is the same sample-accumulator as in §3.

> **Proposal 2 summary.** (a) State: one scalar per layer, the running $\mathbb{E}|h|^2$ and $\mathbb{E}(|h|^2)^2$ (equivalently $\mathrm{Var}(\tilde{\sigma}^2)$). (b) FLOPs: O(n²) per layer. (c) Falsifier: flatness/slope check of stored (2,2) slice vs the single coefficient; per-neuron diagonal corr on |μ/σ|<1.5 set. (d) End-to-end: inject corrected κ_4 diag into mean closure; oracle bound ratio ~0.70 → final MSE ~1.6e−8.

3. Q3 — sampling all three numbers (the unifying estimator)

The mixture structure makes one **control-variate correction sampler** serve all three channels from a single batch of forward passes. Let $g(x)$ be the chain's analytic Gaussian-surrogate prediction for the quantity of interest (variance, κ_4 diagonal, κ_3 diagonal). For each active neuron we estimate the residual $\delta = \mathbb{E}[f] - \mathbb{E}[g]$ where f is the true per-sample statistic:

$$\delta \;\approx\; \frac{1}{N}\sum_{s}\Big[f(x^{(s)}) - g(x^{(s)})\Big], \qquad f \in \big\{(z_i')^2,\ (z_i')^4 - 3((z_i')^2)^2,\ (z_i')^3\big\}.$$

- **Variance (δvar):** need ~1e−3; per-sample residual rms ~1.5e−3 ⇒ N≥9. **N=10³** suffices with 30× margin.
- **κ_4 diagonal:** relative error target ~5–10% of a small value; the mixture coefficient $\mathrm{Var}(\tilde{\sigma}^2)$ is a variance of a squared sum, estimated from the same |h|² accumulators. The κ_4 residual concentrates like the (|h|²) fluctuator; N=10³–10⁴ gives well under the ~20% the chain currently has.
- **κ_3 diagonal:** true value is 0 by symmetry; there is *nothing to sample* — set it to 0 (remove the propagated chain value). This is "infinite variance reduction."

3a. The 100× variance reduction (control variate).

Naïve MC needs ~10^6 samples for variance because it samples the *full* second moment, whose per-neuron spread is ~9% of value — its std is ~0.09, and reducing to 1e−3 needs ~ (0.09/1e−3)² ≈ 8e3... rather the quoted 10^6 counts the per-sample spread across both the quadratic form fluctuation and the closure mismatch. The control-variate version samples only the *residual to the Gaussian surrogate*, whose per-sample std is ~1.5e−3 instead of ~0.09 — a reduction factor $(0.09/0.0015)^2 \approx 3.6\text{e}3$ in samples, i.e. sample count drops from ~10^6 to ~300. **That is the 100× (in fact ~10^3×) cut we wanted.**

3b. Extra cheap structure.

- **1-homogeneity / sphere design:** $f(x) = |x|\, f(x/|x|)$. Draw directions u on the unit sphere (quasi-random / low-discrepancy) and draw radial |x| from the exact 1-D law (chi-tails of the Gaussian). This removes the radial MC noise for free and halves the effective sample variance of the direction-sensitive residuals.
- **Antithetic pairs:** pair u and −u. For the even-statistic corrections (variance, κ_4), the antithetic pair cancels the odd part of the sampling noise. For κ_3 the pairs are exactly degenerate (both give 0), reinforcing the $\delta\kappa_3$=0 conclusion.
- **Rao–Blackwell over the last weight row:** the per-neuron residual $(w_i^\top h)^2$ is a quadratic form in an exactly Gaussian weight row given h, so its conditional-on-h mean is exact: $(w_i^\top h)^2 \to \frac{2}{n}|h|^2$. This is a cheap, exact Rao–Blackwell lift that further collapses the residual noise.

Cost. Total = $0.016\,N$ u for all layers and all three channels. Table:

N samples	Cost	Fraction of budget	δvar residual std (target 5e−4)
300	4.8 u	0.005	8.7e−5 (5.7× under)
1,000	16 u	0.016	4.7e−5 (10× under)
5,000	80 u	0.078	2.1e−5 (24× under)
10,000	160 u	0.156	1.5e−5 (33× under)

Even the aggressive N=10^4 (160 u) is at the leaders' cost ceiling; N=1–5k (16–80 u) exceeds the required accuracy while staying well *below* our current 260 u.

Proposal 3 summary. (a) State: per-neuron accumulators over the active set + running |h|² moments; size O(n). (b) FLOPs: $0.016\,N$ u total; per-layer $\approx (N/1024)\,n^3$. (c) Falsifier: on stored arrays compare the sampled corrections to the stored true δvar, $\delta\kappa_{3iii}$, $\delta\kappa_{4iiii}$; the κ_3 check is "truth≈0, chain value to be removed." (d) End-to-end: inject all three corrected numbers every layer into the mean closure; oracle bound for all-three = ratio 0.057 (truth noise) ⇒ expect final MSE near ~0.13e−8 if every channel is perfect.

4. Q4 — ranking and what to run first

Combine per-channel score gain (from the oracle ratios) with per-unit cost.

Channel	Oracle gain (MSE ratio)	Cheapest sound fix	Cost	Gain per unit (rel.)
variance (−47%)	2.29e−8 → 1.22e−8	CV correction sampler, $N=10^3$	16 u	**very high**
κ_4 diag (−30%)	2.29e−8 → 1.60e−8	mixture coefficient Var(σ^2)	≈0 (no new pass)	highest
κ_3 diag (−14%)	2.29e−8 → 1.97e−8	set to 0 (remove transport)	0	free

Ranking by expected score gain per unit of compute:

1. **κ_3 diagonal (free).** Set the marginal κ_{3iii} to 0 and stop propagating it into the mean closure. Zero cost, worth ~14%. The falsifier (in minutes) confirms the truth is 0 before committing. **Do this immediately — it is reversed in one line of code.**
2. **κ_4 diagonal via the mixture coefficient (≈free).** The scalar $\mathrm{Var}(\tilde{\sigma}^2)$ computed from the existing h-statistics already in the chain gives the level and the (2,2) slice; the finite-n pattern rides the same sampler. This is the highest gain-per-unit.
3. **variance via CV correction sampler (16–80 u).** The only one with real compute cost, but −47% is the largest channel and $N=10^3$ already clears the accuracy bar at 16 u. Runs on the *same* N forward passes as the κ_4 correction.

Experiment to run first: the **κ_3 check** (falsifier 0). It is free, resolves the strongest structural prediction (truth marginal $\kappa_3 = 0$), and if confirmed immediately converts the −14% channel into a dead-simple code change. In the same session, run the **one-batch CV sampler** ($N=10^3$) that simultaneously fixes variance and κ_4 — the same forward passes feed all three corrections — and compare final MSE against the oracle bounds 0.53, 0.70, and 0.057.

5. Consolidated falsifier / end-to-end plan (all under an hour)

```
# F0 (κ₃): regress stored true κ₃(i,i,i) vs chain κ₃ᵢᵢᵢ at each layer.
#         Expect truth ≈ 0 (slope consistent with 0 at MC noise).
# F1 (var): on active set A, v_true = einsum('i,ij,j->i', w_i, C_true, w_i);
#         compare vs chain var and vs CV-sampled var; report max rel err, R².
# F2 (κ₄): flatness/slope of stored κ₄(i,i,j,j) vs Var(σ²) coefficient;
#         per-neuron diag corr on |μ/σ|<1.5 set.
# F3 (end-to-end): inject (δvar, δκ₃=0, δκ₄) into mean closure each layer;
#         re-measure final MSE. Expect ratio ~0.6–0.08 depending on CV quality.
```

Prepared as the Round 7 technical memo. Notes: (i) the Gaussian-mixture law assumes layer-to-layer weight independence, which holds by construction here; (ii) the per-neuron *pattern*, as opposed to the level, is the finite-n weight-row fluctuation on top of the flat mixture level — this is precisely why the chain's own closures cannot predict it and why re-sampling the residual is the correct, cheap tool; (iii) all FLOP counts are in units of n^3 per layer unless stated as total-u forward-pass cost.